Experiment No: 02

Title: Implementation of DML commands of SQL with suitable examples (Insert, Update and Delete)

Topics to be covered:

- Insert table
    - Inserting a single record (way-1)
    - Inserting a single record (way-2)
    - Insert multiple rows
    - Inserting all records from another relation
    - Inserting multiple records
- Update table
- Delete table
    - DELETE-FROM
    - DELETE -FROM-WHERE
    - TRUNCATE

Learning Outcomes:

After completing this experiment, students will be able to:

- Understand the use of DML commands.
- Insert, update, and delete records from a database table.
- Apply SQL commands with correct syntax and structure.

What is DML?

DML (Data Manipulation Language) includes commands used to manipulate data in a database. Common DML commands are:

- INSERT – to add new data
- UPDATE – to modify existing data
- DELETE – to remove data
- TRUNCATE – to remove all data from a table

Consider a table Student:

```
CREATE TABLE Student (
    sno INT,
    sname VARCHAR(50),
    dept VARCHAR(20)
   );
```

1. INSERT Command

The INSERT command adds new rows into a table.

1.1 Inserting a Single Record (Way-1)

Syntax:

```
INSERT INTO table_name VALUES (value1, value2, ..., valueN);
```

Example:

```
INSERT INTO Student VALUES (101, 'Rafiq', 'CSE');
```

1.2 Inserting a Single Record (Way-2)

Syntax:

```
INSERT INTO table_name (column1, column2, ...) VALUES (value1,
value2, ...);
```

Example:

```
INSERT INTO Student (sno, sname) VALUES (102, 'Tania');
```

1.3 Inserting Multiple Rows

Syntax (MySQL):

```
INSERT INTO table_name (column1, column2, ...)
VALUES
(value1, value2, ...),
(value1, value2, ...),
(value1, value2, ...);
```

Example:

```
INSERT INTO Student (sno, sname, dept)
VALUES
(103, 'Hasan', 'CSE'),
(104, 'Amina', 'EEE'),
(105, 'Karim', 'BBA');
```

1.4 Inserting All Records from another Relation

Syntax:

```
INSERT INTO target_table SELECT * FROM source_table;
```

Example:

```
INSERT INTO BackupStudent SELECT * FROM Student;
```

1.5 Inserting Multiple Records (Sub-query or Conditions)

You can insert rows selectively using sub-queries.

Example:

```
INSERT INTO CSE_Students
SELECT * FROM Student WHERE dept = 'CSE';
```

2. UPDATE Command

Used to **modify** existing records in a table.

Syntax:

```
UPDATE table_name
SET column1 = value1, column2 = value2, ...
WHERE condition;
```

Example:

```
UPDATE Student
SET dept = 'CSE'
WHERE sname = 'Amina';
```

3. DELETE Command

Used to **remove records** from a table.

3.1 DELETE FROM (All Records)

Syntax:

```
DELETE FROM table_name;
```

Example:

```
DELETE FROM Student;
```

3.2 DELETE FROM ... WHERE (Specific Records)

Syntax:

```
DELETE FROM table_name WHERE condition;
```

Example:

```
DELETE FROM Student WHERE dept = 'BBA';
```

3.3 TRUNCATE Command (Faster Delete)

Syntax:

```
TRUNCATE TABLE table_name;
```

Example:

```
TRUNCATE TABLE Student;
```

NB: **TRUNCATE** deletes all rows permanently and cannot be rolled back in many databases.

LAB Task-1:

```
CREATE TABLE Student (
    sno INT,
    sname VARCHAR(20),
    dept VARCHAR(10)
);
```

1. Insert the following record into the Student table: (201, 'Rahim', 'CSE')
2. Add a student with name 'Fatema' and sno 202 only (without specifying dept).
3. Insert three students at once:
    - (203, 'Sabbir', 'EEE')
    - (204, 'Kamal', 'CSE')
    - (205, 'Nadia', 'BBA')
4. Copy all records from Student table into a new table named Student_Backup.
5. Create a table CSE_Only, then insert all students of dept ‘CSE’ from Student table.
6. Change the dept of student ‘Kamal’ to ‘BBA’.
7. Update the student name from 'Fatema' to 'Fatima'.
8. Delete all students whose dept is ‘BBA’.
9. Delete all records from Student table using DELETE.
10. Remove all data from Student_Backup using TRUNCATE.

LAB Task-2:

Create the following table:

```
CREATE TABLE Attendance (
    roll_no INT,
    name VARCHAR(30),
    att_date DATE,
    in_time DATETIME
);
```

Tasks:

1. Insert the following record into the Attendance table:
   (101, 'Tuhin', '2025-08-01', '2025-08-01 08:30:00')
2. Insert a new attendance record with roll_no 102 and name 'Lubna' only. (Leave date and time blank/null)
3. Insert three records at once:
   - (103, 'Nayeem', '2025-08-01', '2025-08-01 08:35:00')
   - (104, 'Salma', '2025-08-01', '2025-08-01 08:28:00')
   - (105, 'Nishat', '2025-08-01', '2025-08-01 08:45:00')
4. Create a new table named `Attendance_Backup` and copy all records from `Attendance` to it.
5. Create a new table `Late_Comers`, and insert all students who came after `08:30 AM`.
6. Update the `in_time` of student 'Lubna' to `2025-08-01 08:40:00`.
7. Change the `att_date` of 'Nishat' to `2025-08-02`.
8. Delete all attendance records of students who came after `08:40 AM`.
9. Delete all records from `Attendance` table using `DELETE`.
10. Remove all records from `Attendance_Backup` using `TRUNCATE`.

Tips for Students:

- Use NOW() or 'YYYY-MM-DD HH:MM:SS' format carefully when working with datetime.
- Compare DATETIME using string format or MySQL's TIME() function if needed:
- SELECT * FROM Attendance WHERE TIME(in_time) > '08:30:00';

LAB Task-3:

Create a table EMPLOYEE with following schema:

(Emp_no, E_name, E_address, E_ph_no, Dept_no, Dept_name, Job_id, Salary)

i. Insert at least 5 rows in the table.
ii. Display all the information from EMPLOYEE table.
iii. Display the record of each employee who works in department D10.
iv. Update the city of Emp_no-12 with current city as Nagpur.
v. Display the details of Employee who works in department SALES.
vi. Delete the email_id of employee Ridwan.
vii. Display the complete record of employees working in SALES Department

LAB Task-4:

1. Create the Customers **Table**

- Create a table named **Customers** with the following columns:
  - CustomerID (INT)
  - CustomerName (VARCHAR(100))
  - ContactNumber (VARCHAR(15))
  - Email (VARCHAR(100))

2. Insert Data into the Customers **Table**

- Insert at least three records of customers into the **Customers** table.

3. Update Data in the Customers **Table**

- Update the contact number for 'Alice Johnson'.
- Change the email for 'Bob Smith'.

4. Delete Data from the Customers **Table**

- Remove the record for 'Carol Williams' from the **Customers** table.
- Delete all customer records with CustomerID greater than 2.

5. Create the Meetings **Table**

- Create a table named **Meetings** with the following columns:
  - MeetingID (INT)
  - Subject (VARCHAR(100))
  - MeetingDate (DATE)
  - StartTime (TIME)
  - EndTime (TIME)

6. Insert Data into the Meetings Table

- Insert at least three records into the **Meetings** table with appropriate date and time values.

7. Update Data in the Meetings Table

- Update the subject of a meeting based on its MeetingID.

8. Delete Data from the Meetings Table

- Delete a meeting record based on its MeetingID.